Tropical Plant Biol. (2011) 4:237–249
DOI 10.1007/s12042-011-9085-2

S-RNase-like Sequences in Styles of *Coffea* (Rubiaceae). Evidence for S-RNase Based Gametophytic Self-Incompatibility?

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Received: 16 June 2011 / Accepted: 3 November 2011 / Published online: 17 November 2011

Abstract Although RNase-based self-incompatibility (SI) is suspected to operate in a wide group of plant families, it has been characterized as the molecular genetic basis of SI in only three distantly related families, Solanaceae, Plantaginaceae, and Rosaceae, all described over a decade ago. Previous studies found that gametophytic SI, controlled by a multi-allelic S-locus, operates in the coffee family (Rubiaceae). The molecular genetic basis of this mechanism remains unknown, despite the immense importance of coffee as an agricultural commodity. Here, we isolated ten sequences with features of T2-S-type RNases from two *Coffea* species. While three of the sequences were identified in both species and clearly do not appear to be S-locus products, our data suggest that six sequences may be S-alleles in the self-incompatible *C. canephora*, and one may be a relict in the self-compatible *C. arabica*. We demonstrate that these sequences show style-specific expression, display polymorphism in *C. canephora*, and cluster with S-locus products in a phylogenetic analysis that includes other plant families with RNase-based SI. Although our results are not definitive, in part because the available plant materials were limited and data patterns relatively complex, our results strongly hint that RNase-based SI mechanism operates in the Rubiaceae family.

Keywords *Coffea arabica* · *Coffea canephora* · Rubiaceae · S-RNase · Self-incompatibility

Abbreviations

SI	Self-incompatibility
S-RNase	S-locus ribonuclease

Communicated by Alan Carvalho Andrade

Electronic supplementary material The online version of this article (doi:10.1007/s12042-011-9085-2) contains supplementary material, which is available to authorized users.

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Introduction

Self-incompatibility (SI) in plants is the ability of fertile individuals to recognize and reject self-pollen and prevent inbreeding. Many plant families express some form of SI, which is usually classified by its mode of action (de Nettancourt 1977). In the gametophytic SI (GSI) system, the incompatibility reaction results in the arrest of the pollen tube growth when the haploid pollen carries an allele at the S-locus identical to one of those carried by the diploid pistil. In Plantaginaceae, Solanaceae, and Rosaceae, the pistil products of the S-locus are ribonucleases (S-RNases) (Anderson et al. 1986; Sassa et al. 1993; Xue et al. 1996), but this is not the only existing molecular mechanisms involved, as GSI in *Papaver* is based on entirely different small S-glycoproteins (Thomas et al. 2003).

Crossing studies have previously demonstrated that *Coffea canephora* (Rubiaceae) has monomorphic GSI, controlled by a multi-allelic S-locus (Berthaud 1980; Devreux et al. 1959). In *C. canephora*, the S-locus maps to linkage group nine (Lashermes et al. 1996). However, other species in Rubiaceae are known to express heteromorphic SI (Bawa and Beach 1983), in which different flower morphs are inter-compatible, but identical morphs are not (including those on the same plant). The widespread distribution of RNase-based SI suggests that the pistil-part mechanism of GSI in Rubiaceae may rely on S-RNases. However, the underlying molecular genetic basis of GSI in this family remains unknown, and its elucidation would provide a broader understanding of breeding system evolution in angiosperms.

Most of the approximately 100 *Coffea* species tested for compatibility are reported as SI, with the exception of the polyploid *C. arabica* and the diploid *C. heterocalyx* (Davis et al. 2006; Stoffelen et al. 1996). The relationship between ploidy and SI is the frequent subject of genetic and comparative study (Charlesworth 1985; Mable et al. 2004; Miller and Venable 2000). It is thought that polyploidization disrupts SI system in otherwise diploid species, because of a competitive interaction between different alleles in heteroallelic pollen grains, and this almost invariably occurs in species with RNase-based GSI (Golz et al. 2001, 1999; Huang et al. 2008). A similar event may have occurred in the evolutionary history of *C. arabica*. However, a few polyploid SI species occur in *Prunus* (stone fruits, Rosaceae) (Hauck et al. 2006; Yamane et al. 2001), and consequently, the competitive interaction model may not be universally applicable. The loss of SI is also associated with loss-of-function insertions and deletions, non-synonymous changes, and reduced RNase activity of S-RNases in many diploid species (Stone 2002). A similar mechanism has been invoked as the cause of SC in *C. heterocalyx* (Coulibaly et al. 2002).

The two cultivated species of coffee, *C. arabica* and *C. canephora*, rank among the five most valuable agricultural exports from developing nations (Food and Agriculture Organisation, 2008, from http://www.fao.org). *C. canephora* is generally perceived to provide an inferior flavor, and commands far lower prices in the global commodities markets ($1.22 vs $2.57 per pound, International Coffee Organization, http://www.ico.org/, March 2011), although it provides a number of advantages for its growers over *C. arabica*. Specifically, *C. canephora* seeds contain more caffeine, have higher potential fecundity, its cherries tend not to fall at ripening, and trees can tolerate many pathogens that attack *C. arabica* varieties, as well as a greater range of altitudes and temperatures. But *C. canephora* is SI, and self-fertile plants are both more productive and easier to breed than SI plants.

Therefore, genetic marker-aided breeding could be employed to break down SI in commercial *C. canephora* varieties. Self-compatible *C. canephora* could also be crossed with *C. arabica* to generate hybrid seed for selective breeding. The ability to manipulate SI mechanisms could contribute in improving crop yield and quality, and would provide useful information for breeders and growers to effectively govern pollination. Genetic analyses have already been carried out in other commercially important plants in order to identify fully compatible potential pollinator cultivars and achieve an increased yield (Broothaerts et al. 2004; Goldway et al. 2005; Sapir et al. 2008).

Here, we provide a first glimpse at the mechanism that likely regulates the GSI system in coffee. Below, we report the identification and cloning of putative S-RNases of *C. arabica* and *C. canephora,* and discuss the implications for agronomy and breeding system evolution of flowering plants.

Results

Putative S-RNase Sequencing

We sequenced 112 degenerate primer-amplified RT-PCR products from the sampled mRNA. In *C. canephora*, amplicons were obtained only from stylar cDNAs. Each fragment was ~750 bp long, as expected based on S-RNase sequences from other plant families. The number of independent clones obtained and sequenced for each individual, and the origin of each individual are shown in detail in Table 2. Briefly, we identified four unique transcripts from *C. canephora* Individual Cc1, designated Cc1a, Cc1b, Cc1c and Cc1d. Three unique transcripts were isolated from Individual Cc2, of which Cc2a and Cc2b were novel sequences, not found in Individual Cc1, whereas the third one was identical to Cc1c. Likewise, two novel sequences were isolated from Individual Cc3, named Cc3a and Cc3b, and one novel sequence designated Cc4b from Individual Cc4. In total, we obtained nine unique RNase sequences from *C. canephora* (Table 1).

Only two unique transcripts were found expressed in all varieties of *C. arabica,* after sequencing a total of 62 independent clones. The most abundant clone, Ca1a, was found in ~90% of the sequenced inserts. The remaining six *C. arabica* transcripts with an alternative sequence were identical to the *C. canephora* transcript Cc1c (see Table 2; alignment is shown in Fig. 1). Overall, 10 unique sequences containing an RNase domain were obtained: one sequence from the *C. arabica* samples, eight sequences from the *C. canephora* samples, and one sequence common to both species. Six resulted to be full-length sequences, and four were partial, missing the 5′-end. The alignment of the putative *Coffea* S-RNases amino acid sequences is shown in Fig. 1. Two groups were clearly identifiable in *C. canephora* RNases, showing a low average pairwise identity score (about 20%). The first group, composed of six RNase sequences (Cc1a, Cc2a, Cc2b, Cc3a, Cc3b and Cc4b), shared a long common alignable portion until the last conserved cysteine. The sequence of Cc1a contained some peculiar features such as two insertions (three amino acids each) and one deletion (five amino acids), and a mutation in the highly conserved phenylalanine residue within the C2 domain. The second group was comprised of three proteins, Cc1b, Cc1c and Cc1d, showing a remarkable sequence similarity. A preliminary analysis of raw data from an ongoing *C. canephora* genome sequencing project suggested that Cc1b and Cc1d may have originated from Cc1c through gene duplication events (see "Discussion" for details).

Intron presence and location and pI are both features that are broadly useful in the classification of S/T2-type RNases. In order to empirically determine the number and location of introns in a subset of our sequences, the mRNA and genomic DNA sequences for Ca1a, Cc1b, Cc1c, Cc1d, Cc2a and Cc3a were compared, from the genomic DNA of the *C. arabica* Individual Ca1 (Ca1a) and *C. canephora* Individual Cc3 (Cc3a) and Individual Cc2 (rest). The genomic DNA sequence for Cc4b was obtained from the raw data of an ongoing *C. canephora* genome sequencing project (data not shown). Only one intron was found, located between the C2 and C3 conserved domains of each sequence, except Cc1b. The Cc2a intron was 653 bp, the Cc3a intron was 376 bp, the Ca1a intron was 417 bp, and the Cc4b intron was 2878 bp long. Cc1b, Cc1c and Cc1d shared the canonical S-RNase intron (87 bp; location indicated in Fig. 1). A point mutation in Cc1b introduced a second splice site, leading to a unique second 775 bp intron (site R178, Fig. 1). In addition, the DNA sequence of putative S-alleles (Cc2a, Cc2b, Cc3a, Cc3b and Cc4b) revealed no presence of additional introns, such as those found in class II and III non-S-RNases, or the unique intron observed in Cc1b. A schematic summary of the intron\exon organization and intron length of the sequences analysed is shown in Fig. 2.

Table 1 Sequence-specific primers used in the present work. Reverse primers were used for both 5′-RACE and specificity analysis

Primers	Sequence (5′→3′)
Ca1a_For	ccagataattacacgcaacc
Ca1a_Rev	gaacatgtgccatgacgatt
Cc1a_For	agctggcggaaaatcttatg
Cc1a_Rev	ggctattgaagtcgaatccatc
Cc1b/c/d_For	cttcaccaaaattttgcaaaattg
Cc1b _Rev	cgttgcgtataatccccact
Cc1c/d_Rev	tgaatagtcctgctgaagtatgtagc
Cc2a_For1	tggccygataaytacacgcg
Cc2a_Rev1	yggtacatagattctgcttg
Cc2a_For2	cacctcttttccctccgctc
Cc2a_Rev2	ggtacatagttggagccacgacaa
Cc2b_For	tttcggatgtccaatcggcca
Cc2b_Rev	tgaagacacggagattggct
Cc3a_For	tgctaaatccgccatcagca
Cc3a_Rev	gccacggcaacggaattgga
Cc3b_For	tacgacgtttcgggaagccgagtg
Cc3b_Rev	tcagccttatccgcccaagta
Cc4b_For	agtattgacgctcgagatgac
Cc4b_Rev	atacaataccccgtccagga

All the full length sequences generated in our study had a predicted basic pI, between 8.56 and 10.20 (see Fig. 1 and related caption). S-RNases from other families display a variable, but strongly bounded range, pI=[8.5–10.5]. All Class I RNases show strongly acidic pI=[4.5–6.5], and Class II RNases follow suit with pI=[4.5–6.5], with the exception of one subclade comprised of *Zea mays* kin1 and its homologs, with pI=[8–9] (data not shown).

Expression Analysis

The analysis for tissue-specific expression of the initially obtained sequencing products from *C. canephora* and *C. arabica* (individuals 1 in each case) by RT-PCR revealed that they were only expressed in the pistils of the studied individuals, as expected for S-RNases (Fig. 3), despite other characteristics indicating that they are unlikely to be S-alleles. Primers co-amplifying Cc1c and Cc1d showed that both of these sequences are expressed in each individual of the two species (Fig. 3).

Table 2 Sequences isolated from *Coffea arabica* and *C. canephora*, designated “Ca” and “Cc” respectively. A letter was added to distinguish between different transcript sequences obtained from the same plant. Redundant transcripts identified in more than one plant were named after the first identified sequence in each species

Species	Individual	Transcript name	No. Clones	Sequence name	Non-redundant nomenclature
C.canephora	Cc1	a	7	Cc1a	Cc1a
		b	2	Cc1b	Cc1b
		c	1	Cc1c	Cc1c
		d	1	Cc1d	Cc1d
	Cc2	a	3	Cc2a	Cc2a
		b	2	Cc2b	Cc2b
		c	5	Cc2c	Cc1c
	Cc3	a	13	Cc3a	Cc3a
		b	1	Cc3b	Cc3b
	Cc4	a	11	Cc4a	Cc3a
		b	4	Cc4b	Cc4b
C.arabica	Ca1 (*Laurina*)	a	20	Ca1a	Ca1a
		b	2	Ca1b	Cc1c
	Ca2 (*Sarchimor* x ET-6)	a	12	Ca2a	Ca1a
		b	1	Ca2b	Cc1c
	Ca3 (*Catuai*)	a	8	Ca3a	Ca1a
		b	1	Ca3b	Cc1c
	Ca4 (*Timor* hybrid)	a	16	Ca4a	Ca1a
		b	2	Ca4b	Cc1c

Individual Specificity Analysis

The results of sequence-specific amplifications are summarized in Fig. 4. We obtained products with PCR primers aimed at Cc1b, Cc1c (identical to Ca1b) and Cc1d in all individuals of *C. arabica* and *C. canephora*. Cc1b and Cc1c were amplified in the four additional species tested, but the amplicon obtained in *C. zanguebarie* was shorter than expected for both sequences. The sequence of Ca1a was only found in *C. arabica* and *C. eugenioides*. The remaining sequences were detected in one to three individuals of *C. canephora* from the same geographical region (Fig. 4).

Phylogenetic Tree Construction and Analysis

Phylogenetic analysis divided T2-type RNases into three major classes (Fig. 5). Class I and class II T2-type RNase formed two distinct clusters, while class III RNases formed a highly polymorphic cluster. Within class III, two main clades received significant support. The first consisted of the Rosaceae S-RNases (*Maloideae* and *Prunoideae* in two separate monophyletic clades) along with S-like *Luffa* RNases. The second comprised of the Solanaceae and Plantaginaceae S-RNases, and Rubiaceae sequences. The group of class III T2-type RNases contained both functional and relictual S-RNases (Igic and Kohn 2001; Vieira et al. 2008).

The phylogenetic analysis identified two distinct clades of coffee sequences. *C. canephora* Cc1a, Cc2a, Cc2b, Cc3a, Cc3b, and Cc4b, as well as the *C. arabica* Ca1a and *C. eugenioides* Ce1a formed a well-supported group. Cc1b, Cc1c and Cc1d grouped with a *Hedyotis terminalis* RNase, found in an EST sequencing effort, in a second distinct cluster. The two additional *C. canephora* sequences similarly obtained from a public ESTs database, did not belong to the S-RNase group (class III), according to their phylogenetic placement or sequence features. One belonged to class I, and another one to class II T2-type RNases, which contain proteins of varied functions.

Discussion

Although not definitive, our results introduce several compelling lines of evidence, based on sequence and expression patterns, each of which is consistent with operation of RNase-based SI in Rubiaceae. Specifically, the RNase sequences we obtained were placed in a strongly supported phylogenetic group with active and relictual S-RNases from Solanaceae and Plantaginaceae, which employ RNase-based SI. The presence of a single intron in the position expected for orthologs of S-RNases (class III RNases) solidifies their placement in this clade within the T2/S-RNase superfamily. Each of the putative S-allele sequences was

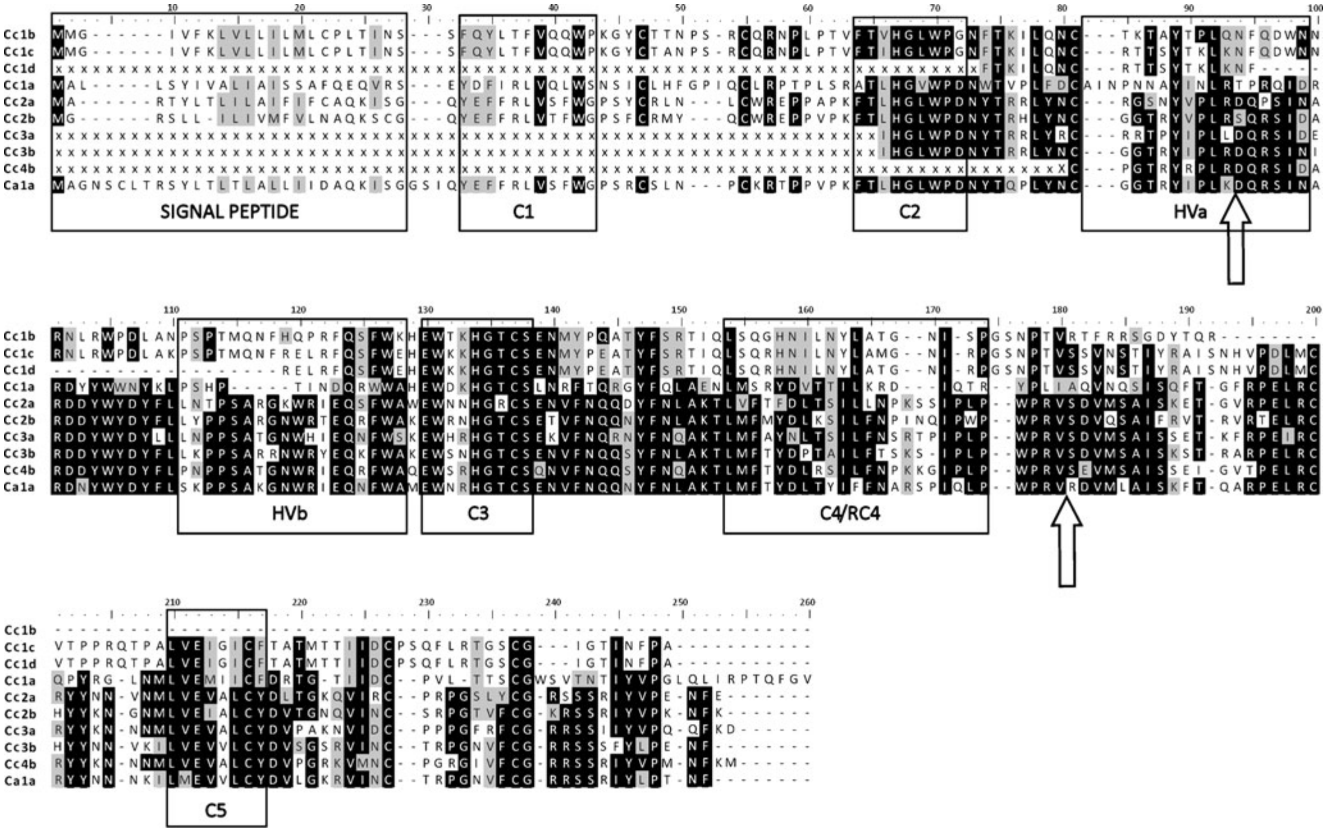


Fig. 1 Amino acid sequences of 10 stylar RNases from *Coffea* identified in this study, aligned using *MUSCLE*. Gaps are encoded as dashes and missing amino acid positions as Xs. Five regions often designated conserved and hypervariable across all S-RNases are labelled (C1, C2, C3, C4/RC4, C5, and HVa/HVb, respectively). The predicted signal peptide is also shown. The location of the canonical S-RNase intron and of the additional intron identified in Cc1b are marked with an arrow. Isoelectric points (concerning the available full length proteins only) are: Cc1b: 10.20; Cc1c: 9.52; Cc1a: 8.56; Cc2a: 8.90; Cc2b: 9.76; Ca1a: 9.85

predicted to have a strongly basic isoelectric point, consistent with those generally observed in functional S-alleles from the three plant families with RNase-based

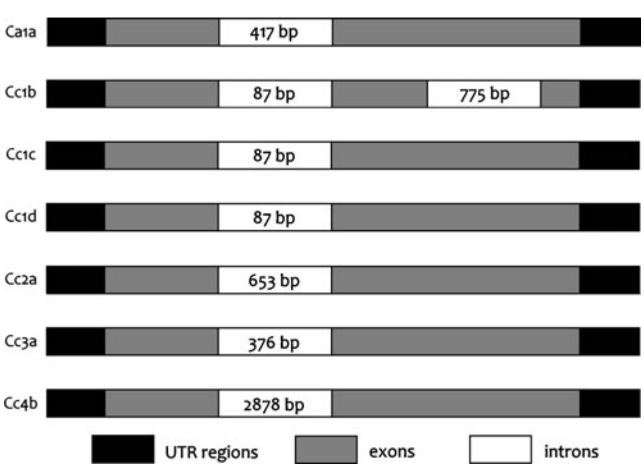


Fig. 2 Schematic summary of the predicted intron\exon organization in the sequences where this feature was analysed (Ca1a, Cc1b, Cc1c, Cc1d, Cc2a, Cc3a and Cc4b). Intron length is shown. For the precise location of intron\exon junctions see the alignment in Fig. 1

SI. Furthermore, six putative S-alleles were found in some, but not all, individuals within *C. canephora*, a feature unlikely to be replicated by non-S-locus genes. These six sequences with individual specificity also displayed a level of sequence divergence comparable to that of S-alleles of other species, had basic isoelectric points, and the intron location (which has only been explored in Cc2b, Cc3b and Cc4b) was consistent with Class III RNases, which are all features strongly associated with S-alleles. Finally, despite our inability to perform controlled crosses due to difficulty in obtaining flowering plants in our greenhouse, the mRNA at hand could be used to show that the putative S-allele transcripts are expressed in pistils, and not stamens. Because the S-RNases act as female components in self-rejection, this observation also supports the hypothesis that the S-RNase-based mechanism is involved in expression of SI in the coffee family. Despite the lack of definitive evidence on the involvement of S-RNases, below we detail our case positing that S-RNase-based SI likely operates in *Coffea* and that it will be difficult to elucidate its detailed mechanism because of the potentially large number of non-functional paralogs.

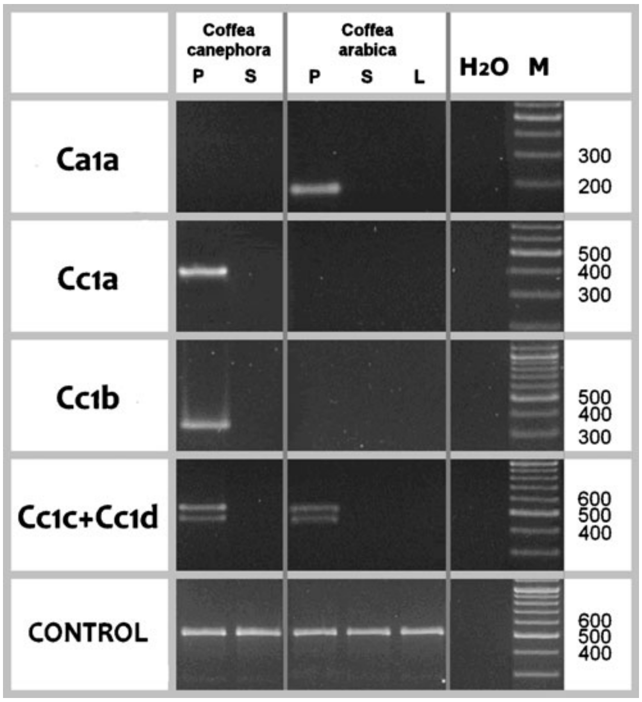


Fig. 3 Expression analyses. Tissue-specific expression of Ca1a, Cc1a, Cc1b, Cc1c (identical to Ca1b) and Cc1d sequences, tested by RT-PCR amplification on *C. arabica* individual Ca1 and *C. canephora* individual Cc1. cDNA quality was evaluated using hexameric polyubiquitin as a control gene. Shorthand labels are: P = pistil, S = stamen, L = leaf; M = 100 bp ladder marker

The phylogenetic analysis (Fig. 5) again recovered the three canonic classes of T2-type RNases (Igic and Kohn 2001; Vieira et al. 2008) and strongly placed the coffee sequences in class III T2-type RNases, highlighting a relationship with Rosaceae, Solanaceae and Plantaginaceae S-RNAses consistent with the APGIII classification . This group contains all S-RNases described so far, as well as relictual RNase genes, which lost their S-activity relatively recently (Golz et al. 1998). However, it also contains several non-S-RNase genes (i.e. *Luffa aegyptiaca* LC1 and LC2), which may have originated from functional S-RNases and subsequently acquired new functions (Igic and Kohn 2001). The sequences of RNases recovered in all individuals of all species of coffee (Cc1b\c\d) may represent precisely such a group, and are unlikely to function in self-rejection. RNases employed in a functional SI system are subject to strong negative frequency-dependent selection and should thus be both heterozygous and found at low frequencies (Wright 1939). Therefore, sequences present in most or all SI individuals are most likely either neo-functionalized copies or pseudogenes that fixed in the population.

Similarly, the sequence designated as Ca1a was found in the genomes of all *C. arabica* (SC) individuals. Nevertheless, it is possible that Ca1a is a relic S-allele, and not a paralogous copy. The S-locus of *C. arabica is* expected to be largely homozygous—depending on the age of breakdown of SI—because it is no longer subject to frequency-dependent selection. In support of its recent function, we found that a diploid congener, *C. eugenioides* (Fig. 4), one of the two proposed parental genomes of *C. arabica*, also harbors a similar putative allele (named Ce1a). Sequencing confirmed that the two sequences are almost identical, with just 3 bp differences across a 660 bp sequence, suggesting

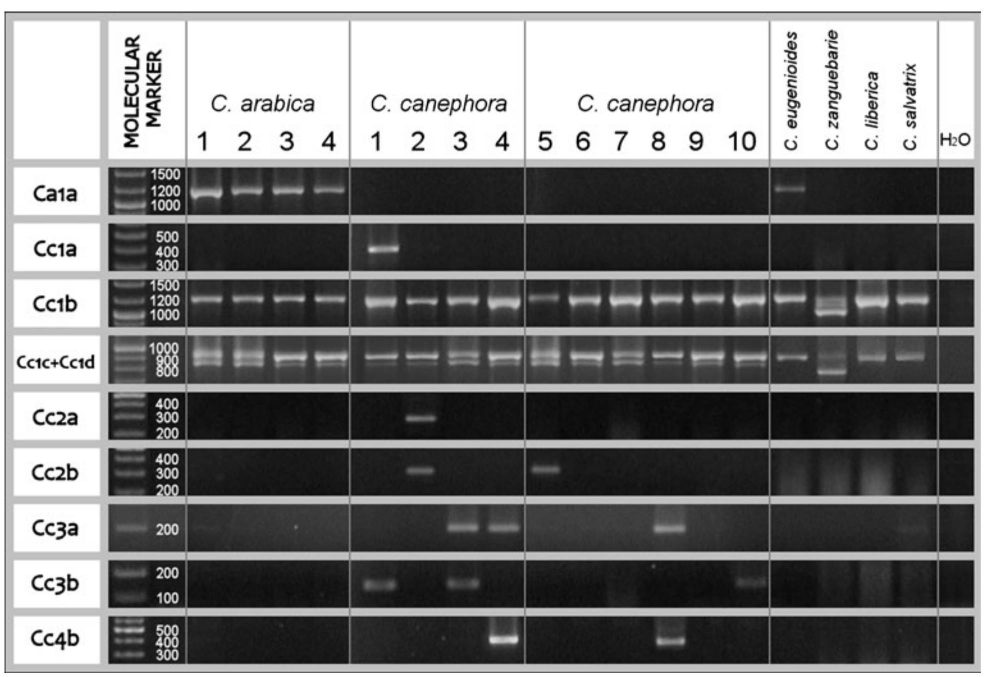


Fig. 4 Sequence-specific amplification of the ten putative S-RNase sequences from *Coffea*. Genomic DNA from each labeled individual, shown in columns, was amplified with primers indicated at the beginning of each row. Primers used for Cc1c (identical to Ca1b) co-amplify Cc1d. Four individuals of both *C. canephora* (Cc1-4) and *C. arabica* (Ca1-4, see Table 2 for details), represent the plants used in the initial sequence isolation step, as detailed in Table 2. We also show amplifications from six additional *C. canephora* individuals (Cc5-10) and four other *Coffea* species (*C. eugenioides*, *C. zanguebarie*, *C. liberica* and *C. salvatrix*)

that Ca1a in cultivated coffee may be derived from *C. eugenioides*, also thought to be one of the wild progenitors of the allopolyploid cultivated species (Lashermes et al. 1999).

The position and absence or presence of introns is a useful heuristic for assessing divergent protein relationships (Garcia-España et al. 2009) including T2/S-type RNases (Igic and Kohn 2001), because their evolution is not as rapid as that of codons. All plant T2/S-type RNases sequenced to date have a plesiomorphic intron, located in the hypervariable region. All sequences that *only* contain this intron belong to Class III RNases (including S-RNases) of Solanaceae, Plantaginaceae and Rosaceae Maloideae (a second intron, located upstream to the hypervariable region characterizes Rosaceae Prunoideae). All the coffee sequences tested for this feature contained this intron, and none of the putative S-allele sequences contained any introns downstream of the hypervariable region (a schematic summary of intron\exon organization of the sequences analyzed is shown in Fig. 2). The sequence Cc1b was characterized by the presence of a second intron, downstream to the HV region, but this is likely the result of a point mutation originating a new splice site in this sequence, highly similar and likely a paralog to Cc1c. In contrast, all members of both Class I and II T2/S-type RNases have other introns downstream, and genes from these groups are not involved in SI (Igic and Kohn 2001).

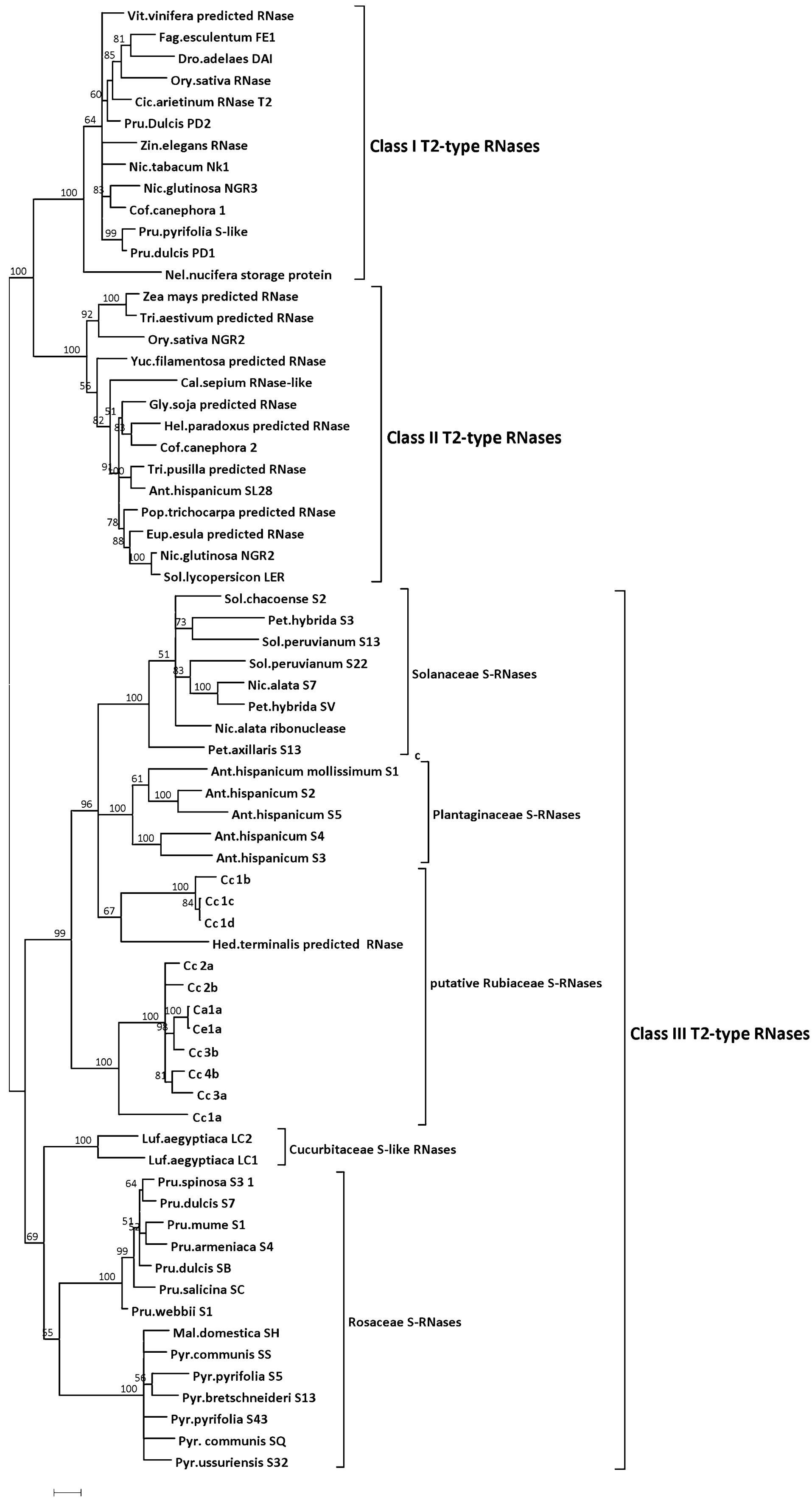
Similarly, isoeletric point (pI), the pH at which a molecule carries no charge, is a rough indicator of the pH for peak enzymatic activity and, by implication, the site of expression or localization. S-RNases are all extracellular proteins with variable size and pI, which enabled their early cloning (Anderson et al. 1989). Nevertheless their pI shows a bounded range, approximately pI=[8.5–10.5]. All Class I RNases show strongly acidic pI=[4–6.5]. With the exception of one subclade comprised of grass species and *Calystegia sepium*, Class II RNases follow suit, pI=[4.5–6.5], with the exceptional clade pI=8–9. Both sets of features, intron position and pI, argue that the phylogenetic grouping of the sequences obtained here is confidently placed with S-RNases, including those from the closely related Solanaceae and Plantaginaceae.

We retrieved two other *C. canephora* T2/S-RNase protein family members from the dbESTs database. Neither is likely to be involved in SI, since both strongly grouped with class I and class II T2-type RNases (Fig. 5), displayed acidic pI and show secondary sequence features consistent with this classification (cysteine location and number) (Vieira et al. 2008). We did not recover either sequence from degenerate PCR amplifications of pistil tissue, and they both have acidic calculated pI (4.70 and 6.28). In short, they very likely belong to non-S-RNase subfamilies of single-copy genes found across nearly all seed plants, generally thought to be involved in wounding response and phosphate harvesting (Bariola and Green 1997).

Phylogenetic analyses of highly divergent and short sequences are complicated by a limited signal to noise ratio, inadequacy of models, and inconsistency of methods (Huelsenbeck and Hillis 1993). Nevertheless, multiple methods and models, as well as data independent of primary sequence alignments (intron position, isoelectric point, interacting loci), all support the interpretation that all RNases acting as female components of SI in Solanaceae, Plantaginaceae, and Rosaceae, are orthologs (molecular homologs) (Igic and Kohn 2001; Steinbachs and Holsinger 2002; Vieira et al. 2008). In the absence of further genetic experiments we cannot determine if the putative alleles are products of the same locus in different individuals, and we are therefore unable to confirm if *C. canephora* sequences are S-alleles. However, the confident placement of the *Coffea* sequences obtained in this study suggests a subset of sequences as good candidates for an involvement in GSI. Additional pattern-based evidence was provided by the sparse distribution among different individuals of the sequences designated Cc1a, Cc2a, Cc2b, Cc3a, Cc3b, and Cc4b.

Our analyses do make it clear that the further genetic experiments and population genetic studies in *Coffea* will be far more complicated than similar studies commonly performed in Solanaceae, as the copies of recently derived S-allele paralogs could represent substantial hurdles for research in elucidation of SI mechanism based on sequencing approaches.

Our expression analyses were impeded by lack of tissue availability, which resulted in experiments performed on transcripts that are not likely to be functional S-alleles. Nevertheless, the phylogenetic proximity of these transcripts to putative S-alleles and their pattern of expression indicated that they genes may have only recently duplicated and diverged from true S-alleles. In SI *C. canephora* Cc1a, Cc1b, Cc1c, and Cc1d are all expressed in pistils, but not in stamens (see Fig. 3). It remains unclear whether any of the three sequences Cc1b, Cc1c and Cc1d, whose pattern of expression is retained, and which are found within the genomes of many, if not all, species tested (see Fig. 4), has retained an active RNase function in an unrelated process. A preliminary analysis of raw data from an ongoing *C. canephora* genome sequencing project revealed that multiple Cc1c gene copies are present in the coffee genome (data not shown), and therefore both Cc1d (different from Cc1c just for a 60 bp deletion) and Cc1b (where a point mutation leads to the introduction of a second intron) may have originated from Cc1c through gene duplication events. On the contrary, Cc1a retains individual specificity (see Fig. 4), as expected for a functional S-allele, but some peculiar


Vit.vinifera predicted RNase
Fag.esculentum FE1
Dro.adelaes DAI
Ory.sativa RNase
Cic.arietinum RNase T2
Pru.Dulcis PD2
Zin.elegans RNase
Nic.tabacum Nk1
Nic.glutinosa NGR3
Cof.canephora 1
Pru.pyrifolia S-like
Pru.dulcis PD1
Nel.nucifera storage protein
Class I T2-type RNases
Zea mays predicted RNase
Tri.aestivum predicted RNase
Ory.sativa NGR2
Yuc.filamentosa predicted RNase
Cal.sepium RNase-like
Gly.soja predicted RNase
Hel.paradoxus predicted RNase
Cof.canephora 2
Tri.pusilla predicted RNase
Ant.hispanicum SL28
Pop.trichocarpa predicted RNase
Eup.esula predicted RNase
Nic.glutinosa NGR2
Sol.lycopersicon LER
Class II T2-type RNases
Sol.chacoense S2
Pet.hybrida S3
Sol.peruvianum S13
Sol.peruvianum S22
Nic.alata S7
Pet.hybrida SV
Nic.alata ribonuclease
Pet.axillaris S13
Solanaceae S-RNases
Ant.hispanicum mollissimum S1
Ant.hispanicum S2
Ant.hispanicum S5
Ant.hispanicum S4
Ant.hispanicum S3
Plantaginaceae S-RNases
Cc1b
Cc1c
Cc1d
Hed.terminalis predicted RNase
Cc2a
Cc2b
Ca1a
Ce1a
Cc3b
Cc4b
Cc3a
Cc1a
putative Rubiaceae S-RNases
Class III T2-type RNases
Luf.aegyptiaca LC2
Luf.aegyptiaca LC1
Cucurbitaceae S-like RNases
Pru.spinosa S3 1
Pru.dulcis S7
Pru.mume S1
Pru.armeniaca S4
Pru.dulcis SB
Pru.salicina SC
Pru.webbii S1
Mal.domestica SH
Pyr.communis SS
Pyr.pyrifolia S5
Pyr.bretschneideri S13
Pyr.pyrifolia S43
Pyr. communis SQ
Pyr.ussuriensis S32
Rosaceae S-RNases
0.2

◀ **Fig. 5** Phylogeny of plant T2/S-RNase family. Posterior probabilities derived from the Bayesian analysis are shown above the branches of the tree. Sequences obtained in the present study retain their assigned names (see Table 2, Figs. 1 and 3). S-RNases are involved in self-incompatibility in three families, indicated on the right, and implicated in Rubiaceae. Names of genera are abbreviated as follows: Pet., *Petunia*; Sol., *Solanum*; Ant., *Antirrhinum*; Hed., *Hedyotis*; Nic., *Nicotiana*; Pyr., *Pyrus*; Pru., *Prunus*, Mal., *Malus*; Luf., *Luffa*; Cal., *Calystegia*; Cof, *Coffea*; Gly., *Glycine*; Pop., *Populus*; Tip., *Tiphysaria*; Ory., *Oryza*; Tri., *Triticum*; Dro., *Drosera*; Fag., *Fagopyrum*; Zin., *Zinnia*; Vit., *Vitis*; Cic., *Cicer*; Hor., *Hordeum*; Nel., *Nelumbo*; Yuc., *Yucca*; Hel., *Heliantus*; Eup., *Euphorbia*

sequence features, in particular the substitution of the highly conserved Phe with Ala within the conserved domain C2 raise questions about its effective involvement in SI.

In the SC *C. arabica,* Ca1a, though clearly not a presently part of an active SI system, is expressed in pistils, not stamens, perhaps as a relictual remnant of the S-locus. A similar pattern is found in the SC *Nicotiana sylvestris*, where a functional, yet not involved in SI stylar S-RNase can be identified (Golz et al. 1998). Bolstering the case as a possible relic, we also found a close relative of Ca1a in the progenitor, the SI *C. eugenioides* (Ce1a; Fig. 4), one of the two likely parental genomes of allotetraploid cultivated *C. arabica* (Lashermes et al. 1999). The two sequences are almost identical with just three nucleotide differences over 660 bp. The second sequence identified in *C. arabica*, Ca1b, was identical to Cc1c and amplified from all individuals and species tested, along with Cc1b and Cc1d, whose presence in this species was not detected using degenerated primers, but was highlighted by using sequence specific primers from genomic DNA (see Fig. 4). For this analysis, Ca1b/Cc1c was amplified from two *C. arabica* cv. *Laurina* and *Catuai* individuals and two hybrids with extensive *C. canephora* introgressions (cv. *Timor* hybrid and a wild plant ET-6 x *Sarchimor* cross), each self-fertile (Lashermes et al. 2000). The hybrids appear to have inherited the cv. *Arabica* S-locus, given that we could not find any RNases distinct from other *C. arabica* cultivars. This is an indication of low diversity at the S-locus in *C. arabica*, which may be unsurprising, given that the cultivated variety was obtained through several rounds of backcrosses during which the breeders were almost certainly selecting for the most self-fertile plants.

Although many aspects of our study lack definitive clarity, the results provide compelling evidence that RNases may be involved in SI reactions of *Coffea* species, long known to express the gametophytic system (Devreux et al. 1959). If so, the distribution of SI will have been confirmed in three orders of the species-rich euasterid I clade (APGII, 2003). Therefore this information could forge a path for further inquiry into the S-RNase SI mechanism, not only in Rubiaceae but also in other Asteridae, including many herbaceous species with a described GSI mechanism (Igic et al. 2008), amenable to genetic research, especially with accessibility to next generation sequencing techniques.

Our findings directly suggest follow-up studies to ascertain the exact identity of the S-locus in Rubiaceae. Although appropriate material is difficult to obtain, very few flowering individuals can be used for controlled crosses, to track allele segregation, and expression analyses from multiple tissues. In addition, both commercial and government-sponsored entities are pursuing whole-genome sequencing of coffee. The complete genome sequence will enable easier localization of the S-locus, at least because of the predicted pairing of RNases and with F-boxes at a candidate (Qiao et al. 2004; Sijacic et al. 2004; Ushijima et al. 2003).

Methods

Plant Material

Pistils and stamens of *C. canephora* cv. S274 from four different individuals were provided by the Coffee Board of India. They were collected in the district of Dindigul, Tamil Nadu, and conserved in *RNAlater*® solution (Applied Biosystems/Ambion, Austin, Texas) to preserve RNA integrity until extraction. Matching accession leaf samples were available for DNA extraction, but not for RNA extraction, because they were air-dried and not appropriately preserved. The limited amount of starting material prevented some necessary analyses, as described below. Unfortunately, the expression analyses could only be performed on a small amount of available mRNA (see "Expression Analyses" below). Flowers and leaves from four different *C. arabica* plants (*Laurina*, *Catuai, Timor* hybrid varieties, and one F_2 plant obtained from a wild plant ET-6 x *Sarchimor* cross) were collected from the greenhouse of the Life Sciences Department at the University of Trieste. The same university's germplasm collection yielded the DNA samples used for the sequence-specific genotyping analyses.

Amplification and Sequencing

Total RNA was isolated from styles and stamens of flowers, collected 1–2 days before anthesis, as well as leaves of *C. arabica,* following the protocol of Chang et al. (1993). First strand cDNAs were generated from 1 μg of total RNA with Superscript II RNase H-Reverse Transcriptase (Invitrogen) and an oligo-dT adapter primer (attB2_oligodT: 5′-biotin-GGGGACCACTTTGTACAA GAAAGCTGGGCGGCCGC$(T)_{20}$VN-3′) in the presence of SMART_T7 primer (5′-ACTCTAATACGACTCACTA TAGGGArGrGrG-3′).

For 3′-RACE PCR (Frohman et al. 1988) on these cDNAs a degenerate primer designed to target the conserved region C2, and a primer complementary to the added 3′ adaptor sequence (attB2: 5′-GGGGAC CACTTTGTACAAGAAAGCTGGG-3′) were used.

The degenerate primer complemented a consensus of S-RNase sequences deposited in public databases from species close to Rubiaceae. Several GenBank accessions representative of Solanaceae (X03803.1, AF232304.1, X76065.1, AF176533.1, U28795.1, U28796.1 and S69589.1), and Plantaginaceae S-RNases (X96464.1 and X96466.1) were aligned using CLUSTALW software (Thompson et al. 1994) and used to generate a degenerate oligonucleotide forward primer used in PCR amplification (C2For; 5′-TTYACVATYCAYGGGCTHTGGC-3′; Y means C or T; H means A or C or T; V means A or C or G).

PCRs were performed using a temperature profile with an initial denaturing step at 94°C for 3 min, followed by 33 cycles at 94°C for 30″, 60°C for 30″, 72°C for 1′30″, and a final extension step at 72°C for 10′. PCR products were then separated in a 0.8% (w/v) agarose gel, stained with ethidium bromide. The products were purified using MinElute Gel Extraction Kit (Qiagen, Venio, Netherlands) and cloned using CloneJET cloning kit (Fermentas, Glen Burnie, Maryland). The cloned inserts were sequenced from several individual bacterial colonies and redundant sequences were identified by a comparative sequence analysis.

Allele-specific primers were designed to obtain full-length transcripts of *C. canephora* individuals Cc1 and Cc2, and of the *C. arabica* transcript Ca1a by 5′-RACE (Table 1). 5′-RACE was performed following the same protocol used for 3′-RACE, using allele-specific primers, and the primer 12AttB1_T7 (5′-AAAAAGCAGGCTAATACGACTCAC TATAGGGA-3′). Sequences were assembled using the *Seqman* application included in the *Lasergene* software package (DNAstar Inc., Madison, Wisconsin).

The *C. arabica* and *C. canephora* sequences were designated AR and CA, respectively. A number suffix was used to indicate the individual from which the sequence was extracted, and a letter was added to distinguish among different sequences obtained from the same individual. The acquired sequences were deposited in EMBL Nucleotide Sequence Database (EMBL Nucleotide Sequence Database accession IDs: FN547910-FN547919).

Because the presence of only one intron, located between the conserved domains C2 and C3, is a canonical feature of S-RNases in euasterid families (Igic and Kohn 2001), we searched for the presence of introns and checked for their location, which was inferred from the subtraction of cDNA from genomic DNA sequences. The primer pairs Cc2aFor1/Cc2aRev1, Cc1b-c-dFOR/Cc1bREV, Cc1b-c-dFor/Cc1c-dRev, C2For/Cc3aRev and Ca1aFor/Ca1aRev (Table 1) were used to determine intron location and length of Cc1b, Cc1c, Cc1d, Cc2a, Cc3a and Ca1a by the amplification of *C. canephora* individual Cc2 and Cc3 and *C. arabica* Individual Ca1 (*Laurina*) genomic DNA. Length and location of the intron in Cc4b were inferred from the raw data obtained from an ongoing *C. canephora* genome sequencing effort (data not shown).

A basic isoelectric point (pI) is a conserved feature among functional S-RNases. Isoelectric points (pI) of the available full length predicted proteins were estimated using the calculator implemented in *seqinr* package of the *R* programming language (Charif and Lobry 2007).

Expression Analyses

Total RNA was extracted and treated with 1 μg of DNase I (Applied Biosystems/Ambion, Austin, Texas) to avoid contamination by genomic DNA. The extracted RNA was then reverse-transcribed in 30 μl reaction mix with 22.5 units of MultiScribe Reverse Transcriptase and 1.25 μmol/L Oligo-d(T)16, according to the manufacturer's protocol (GeneAmp Gold RNA PCR Core Kit, Applied Biosystems, Austin, Texas). A PCR control reaction without the reverse transcriptase (no-RT) was performed to check for genomic contamination. Sequence-specific primers (Table 1) were designed to test the expression of Ca1a, Cc1a, Cc1b, Cc1c (identical to Ca1b), and Cc1d with RT-PCR in different tissues of *C. arabica* Individual Ca1 and *C. canephora* Individual Cc1. Ca1b/Cc1c and Cc1d share common sequence-specific primers. The hexameric polyubiquitin (GenBank accession AM292305), a housekeeping gene expressed in all tissues, was used as a positive control for cDNA presence and quality (De Nardi et al. 2006; Fernandez et al. 2004). Primers used for its amplification were polyUb_For: GCAGATTTTCGTCAAGACATTGACTG; polyUb_Rev: ACCAAGGCAAGTTCAGTTCAACAC.

RT-PCR could only be performed on a limited amount of available material. Consequently, we were unable to perform additional analyses using allele-specific primers to determine the expression pattern of the five putative S-alleles (Cc2a, Cc2b, Cc3a, Cc3b, and Cc4b), obtained after the initially determined sequences (Ca1a, Cc1a, Cc1b, Cc1c (identical to Ca1b), and Cc1d; see "Discussion" for details).

Genotyping

With the aim of establishing the connection between the novel coffee sequences and true S-RNase, we designed sequence-specific primer pairs (Table 1). Despite their stylar expression, these sequences could be the product of non-S-allele genes, which could be present in most (if not all) individuals of a species, and possibly in other closely related species. Conversely, true S-RNases are expected only in a limited number of individuals. Besides the four *C. canephora* and *C. arabica* individuals, PCR reactions were also carried out on genomic DNAs of six additional canephora S274 individuals from the same geographical region, and four other *Coffea* species (*C. liberica*, *C. salvatrix*, *C. eugenioides* and *C. zanguebarie*). PCR reactions were performed using a temperature profile with the initial denaturing step at 94°C for 1′00″; 35 cycles of 94°C for 20″, 58°C for 1′30″, 72°C for 1′00″, and the final extension step at 72°C for 10′. Amplified fragments were separated on 2% agarose gel, and visualized by staining with ethidium bromide. Some amplicons were then sequenced to confirm specificity of amplification.

Phylogenetic Tree Construction and Analysis

We used an alignment of our putative S-allele sequences along with a representative sample of other full-length T2-type RNases, as this protein family is known, to reconstruct their phylogenetic relationships and assess the position of our *Coffea* sequences, relative to known S-RNases. We identified a sample of suitable sequences from a BLASTp search of GenPept non-redundant (*nr*) database, using the BLOSUM62 cost matrix, with gap opening and extension penalty 10 and 1 respectively, with full length plant T2-type RNases, designated in classes I-III .

In the final analysis, 12 sequences were selected from Class I (non-S-RNases), six from Class II (non-S-RNases), and 29 from Class III (mainly S-RNases; 14 from Rosaceae, eight from Solanaceae, five from Plantaginaceae, and two non-S RNases from Cucurbitaceae). Since only a few complete amino acid sequences were retrieved for Class II RNases, this group was enriched with seven additional putative S-like sequences obtained from ESTs, detected with a tBLASTn search in the GenBank *est_others* database, using known Class II RNase queries. Finally, other putative S-RNase sequences from Rubiaceae were used as search templates for tBLASTn method in *est_others* GenBank database. These searches yielded two additional *C. canephora* full-length putative S-like proteins (DV711148, DV676616 and EE193267, DV691178, DV691510) and an incomplete *Hedyotis terminalis* sequence (CB078110).

Our sequences (Cc1a,Cc1b, Cc1c, Cc1d, Cc2a, Cc2b, Cc3a, Cc3b, Cc4b, Ca1a and its *C. eugenioides* homologue Ce1a) were added to these database accessions, and the entire dataset was aligned using *MUSCLE* (Edgar 2004). The alignment was manually refined in *MEGA v5.0* (Kumar et al. 1994). The sequence data of the signal peptide as well as the region following the last conserved cysteine was discarded, due to the high divergence and alignment difficulty, as in Igic and Kohn (2001). The GenBank\GenPept accession IDs of all the sequences used for the analysis are listed in Table S1 (Supplementary Material).

In order to reconstruct phylogenetic relationships among the obtained sequences, we used the Bayesian inference, which relied on a Metropolis-coupled Markov chain Monte Carlo sampling method, implemented in *MrBayes* v 3.1 (Huelsenbeck and Ronquist 2001). The *GTR* substitution model of molecular evolution with an estimated proportion of invariable sites, and a Gamma-shaped distribution of rates across sites ($GTR + \gamma + I$), was chosen as the best-fitting model for our dataset (Abascal et al. 2005). We ran two independent analyses with four chains each (one “cold”, three “warm”), for five million generations, sampling a single tree each 50,000 generations. *Tracer v*1.4 (Rambaut and Drummond 2007) was used to visually inspect convergence of the samples obtained from the posterior distribution. The first 20 trees were discarded for the burn-in procedure, and the remaining trees were used to calculate the posterior probability for each node in a 50% consensus tree.

Acknowledgements We thank Dott. Manoj Kumar Mishra and the Coffee Board of India for providing part the plant material, and Paolo Zago for plant care. We also thank the anonymous reviewers for their precious suggestions which provided a great help for the improvement of the paper.

The GenBank\GenPept accession IDs of all the sequences used for the analysis are listed in Table S1 (Supplementary Material).

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